6 **(a)** **(i)** $BI\sin\theta$ .... B1 [1]

(ii) (downwards) into (the plane of) the paper .... B1 [1]

(b) **(i)** magnetic field (due to current) in one loop OR each loop acts as a coil B1
cuts/is normal to current in second loop OR produces magnetic field B1
causing force on second loop OR fields in same direction M1
either Newton's 3rd discussed
or vice versa clear gives rise to attraction OR so attracts .... A1 [4]

(ii) $B = 2 \times 10^{-7}\, I/0.75 \times 10^{-2}$ ($= 2.67 \times 10^{-5}\, I$) C1
force $= 0.26 \times 10^{-3} \times 9.81$ ($= 2.55 \times 10^{-3}$ N) C1
$F = BIL$
$2.55 \times 10^{-3} = 2.67 \times 10^{-5} \times I^2 \times 2\pi \times 4.7 \times 10^{-2}$ C1
$I = 18$ A .... A1 [4]

7 (a) energy required to (completely) separate the nucleons (in a nucleus) B1 [1]